Implement a vehicle detection algorithm using Open CV to detect and locate vehicles in each frame of the video.

Code:

```
import cv2, urllib.request

from google.colab.patches import cv2_imshow

# Download sample video

urllib.request.urlretrieve(

    "https://raw.githubusercontent.com/opencv/opencv/master/samples/data/vtest.avi",

    "video.avi"

)

# Download Haar Cascade for cars

urllib.request.urlretrieve(
"https://raw.githubusercontent.com/andrewssobral/vehicle_detection_haarcascades/master/cars.xml",

    "cars.xml"

)

# Load classifier

car_cascade = cv2.CascadeClassifier("cars.xml")

# Read video

cap = cv2.VideoCapture("video.avi")

while cap.isOpened():

    ret, frame = cap.read()

    if not ret:

        break

    # Convert to grayscale

    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

    # Detect vehicles

    cars = car_cascade.detectMultiScale(gray, 1.1, 2)

    # Draw rectangles

    for (x, y, w, h) in cars:

        cv2.rectangle(frame, (x,y), (x+w,y+h), (0,255,0), 2)
```

```
    # Display result
    cv2_imshow(frame)
    if cv2.waitKey(30) & 0xFF == 27:
        break
cap.release()
```

Output: